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# AIPatient: Simulating Patients with EHRs and LLM Powered Agentic Workflow

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## Abstract

Simulated patient systems play a crucial role in modern medical education and research, providing safe, integrative learning environments and enabling clinical decision-making simulations. Large Language Models (LLM) could advance simulated patient systems by replicating medical conditions and patient-doctor interactions with high fidelity and low cost. However, ensuring the effectiveness and trustworthiness of these systems remains a challenge, as they require a large, diverse, and precise patient knowledgebase, along with a robust and stable knowledge diffusion to users. Here, we developed **AIPatient**, an advanced simulated patient system with AIPatient Knowledge Graph (**AIPatient KG**) as the input and the Reasoning Retrieval-Augmented Generation (**Reasoning RAG**) agentic workflow as the generation backbone. AIPatient KG samples data from Electronic Health Records (EHRs) in the Medical Information Mart for Intensive Care (MIMIC)-III database, producing a clinically diverse and relevant cohort of 1,495 patients with high knowledgebase validity (F1 0.89). Reasoning RAG leverages six LLM powered agents spanning tasks including retrieval, KG query generation, abstraction, checker, rewrite, and summarization. This agentic framework reaches an overall accuracy of 94.15% in EHR-based medical Question Answering (QA), outperforming benchmarks that use either no agent or only partial agent integration. Our system also presents high readability (median Flesch Reading Ease 77.23; median Flesch Kincaid Grade 5.6), robustness (ANOVA F-value 0.6126,  $p>0.1$ ), and stability (ANOVA F-value 0.782,  $p>0.1$ ). The promising performance of the AIPatient system highlights its potential to support a wide range of applications, including medical education, model evaluation, and system integration.

**Keywords:** Simulated Patient, Large Language Models, Electronic Health Records, Multi-agent Systems, Retrieval Augmented Generation, Medical QA, Knowledge Graph

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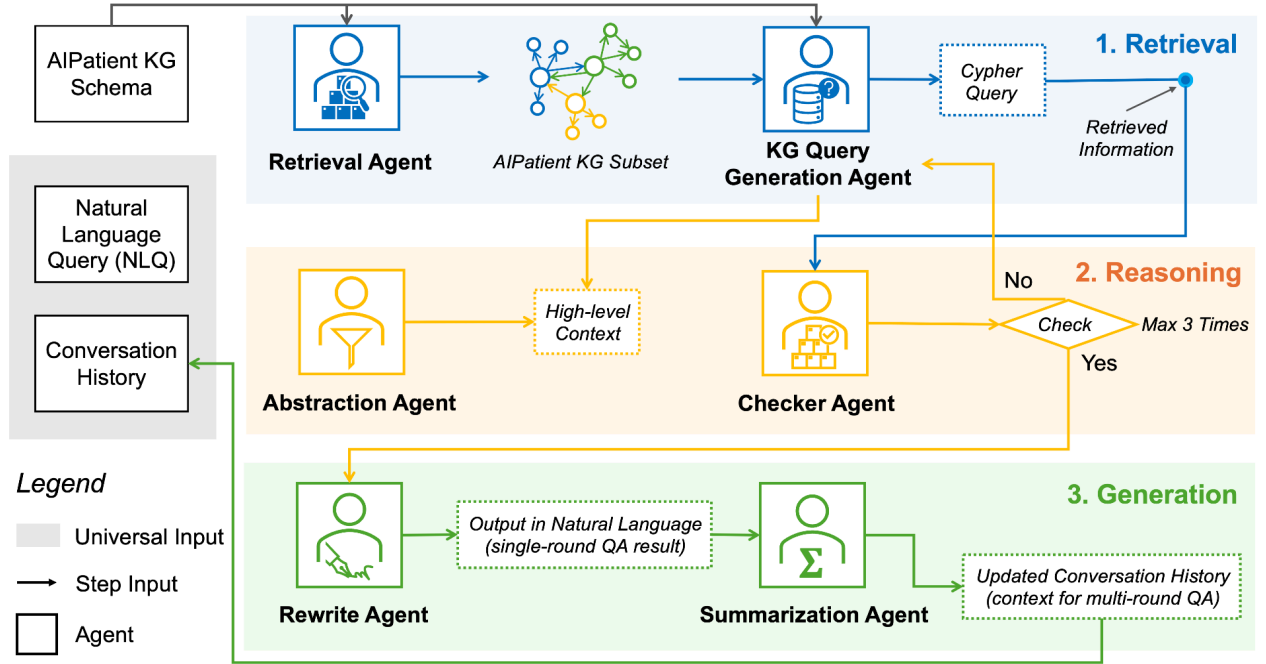
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# Introduction



**Figure 1.** Reasoning RAG agentic workflow is the AIPatient system’s processing backbone, comprising three key stages: retrieval, reasoning, and generation. It first retrieves relevant information from the knowledge graph, then applies contextual reasoning to reduce hallucinations, and finally generates natural language responses based on conversation continuity and tailored to the perceived patient personality.

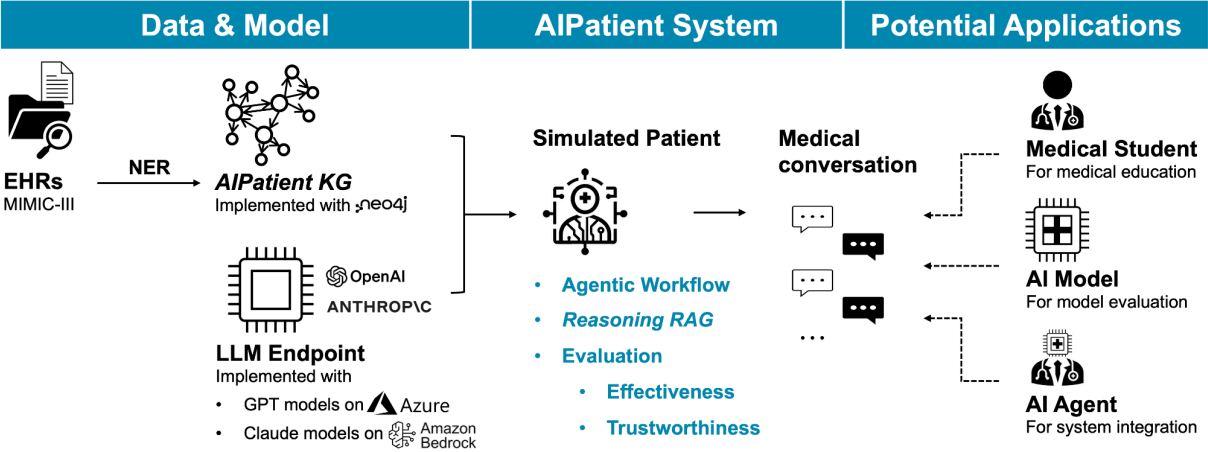
Simulated patient systems, also referred to as virtual patient simulation systems, have become an essential tool in modern medical education and research [1]. These systems are designed to enhance integrative learning and evaluation by incorporating basic science objectives, simulating the outcomes of clinical decisions, and including diverse cases to improve cultural competency [2–5]. The applications of simulated patients are broad, spanning learning, teaching, assessment, and implementation [6,7].

While simulated patients are designed to be a low-risk and high-fidelity tool, there are unresolved concerns about its effectiveness and trustworthiness. Current usability evaluation of simulated patients is usually objective [8]. As a result, for example, variations of medical students’ familiarity of simulated patient systems can also cause inaccuracy in SP-based medical skill evaluation [9]. Moreover, traditional evaluation of simulated patient systems often focuses more on the fact-based accuracy and utilizes inter-rater reliability-based evaluation methods that are less generalizable [10]. For instance, in psychiatric education, inherent issues with utilizing simulated patients are highlighted in connection to the essence of empathy and the distinctively interpersonal nature of the field [11], which are traditionally not the focus of simulated patient system evaluation.

Large Language Models (LLM) are powerful AI systems [12–14], especially renowned for their advanced reasoning capabilities [15–17] and medical applications [18–21]. LLM-based

simulated patients have great potential in consistently replicating a wider range of medical conditions, simulating patient interactions, and mitigating logistical challenges associated with using human actors. Some researchers have developed frameworks and methodologies for general medical education settings [22–26], while others have tailored these technologies to specific use cases such as psychiatry [27,28]. While these implementations are promising, there are still critical gaps in effectiveness and trustworthiness, including (1) the lack of a large, comprehensive, and diverse patient profile database, (2) the need for fact-checking mechanisms to minimize hallucinations and ensure consistency, (3) the insufficient flexibility in taking on different personalities, and (4) the absence of a thorough evaluation framework.

Here, we introduce **AIPatient**, an advanced simulated patient system with LLM-powered *Reasoning Retrieval-Augmented Generation* (Reasoning RAG) agentic workflow and reliable knowledge input from an *AIPatient Knowledge Graph* (AIPatient KG). **Figure 1** illustrates the Reasoning RAG agentic workflow which improves the traditional RAG framework [29], inserting step-by-step reasoning in between. **Figure 2** presents an overview of the AIPatient system, illustrating how it functions as an integrated platform. The system processes realistic data and delivers verified information in a natural language format tailored to the user's needs, particularly aligning responses with patient personalities. Additionally, it ensures continuity in the interaction by summarizing and updating the conversation history throughout the process. Our AI Patient system could benefit medical education and research applications, such as serving as simulated patients in medical student education, facilitating patient-focused evaluation on AI models, and integrating as a patient agent in multi-agent AI systems.



**Figure 2:** AIPatient as an integrative system with EHR-based knowledge input, LLM endpoints as preprocessing and reasoning models, and potential applications.

# Methods

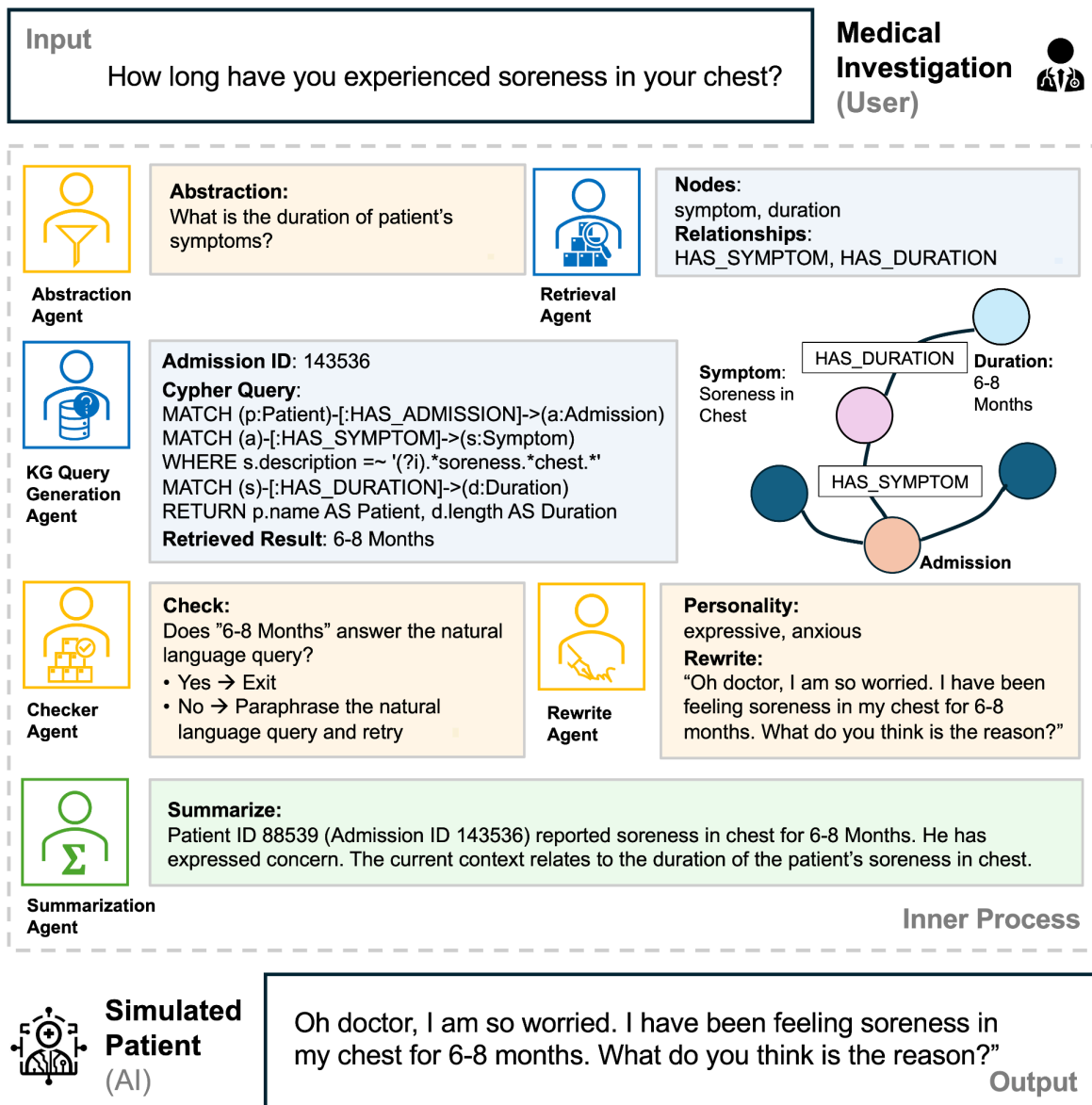
## Data

We create the patient profiles based on 1,500 sampled patient records from the Medical Information Mart for Intensive Care (MIMIC)-III database [30], containing de-identified EHRs for more than 40,000 distinct hospital admissions to critical care units at the Beth Israel Deaconess Medical Center between 2001 and 2012. **Table 1** describes the demographic and clinical characteristics of patient populations in this study. This data use is approved by the PhysioNet Clinical Databases for credentialed access.

| <b>Table 1</b> Patient demographics and clinical characteristics |                           |                         |              |
|------------------------------------------------------------------|---------------------------|-------------------------|--------------|
| Medical Entity                                                   | Attributes                | Categories              | Overall      |
| Patient                                                          | Number of Patients        |                         | 1500         |
|                                                                  | GENDER, n (%)             | F                       | 608 (40.5)   |
|                                                                  |                           | M                       | 892 (59.5)   |
|                                                                  | AGE, mean (SD)            |                         | 51.0 (18.3)  |
|                                                                  | ETHNICITY, n (%)          | Asian                   | 55 (3.7)     |
|                                                                  |                           | Black/African American  | 128 (8.5)    |
|                                                                  |                           | Hispanic/Latino         | 71 (4.7)     |
|                                                                  |                           | Other                   | 252 (16.8)   |
|                                                                  |                           | White                   | 994 (66.3)   |
|                                                                  | RELIGION, n (%)           | Christian               | 592 (39.5)   |
|                                                                  |                           | Non-Christian Religions | 88 (5.9)     |
|                                                                  |                           | Other/Unspecified       | 820 (54.7)   |
|                                                                  | MARITAL_STATUS, n (%)     | Married                 | 742 (49.5)   |
|                                                                  |                           | Other/Unspecified       | 128 (8.5)    |
| Admissions                                                       |                           | Single                  | 630 (42.0)   |
|                                                                  | DURATION, mean (SD)       |                         | 4.5 (4.6)    |
|                                                                  | ADMISSION_TYPE, n (%)     | Emergency               | 1,435 (95.7) |
|                                                                  |                           | Urgent                  | 65 (4.3)     |
|                                                                  |                           | Emergency Admission     | 916 (61.1)   |
|                                                                  | ADMISSION_LOCATION, n (%) | Referral                | 296 (19.7)   |
|                                                                  |                           | Transfer                | 288 (19.2)   |
|                                                                  |                           | Facility                | 172 (11.5)   |
|                                                                  |                           | Home                    | 1,183 (78.9) |
|                                                                  | DISCHARGE_LOCATION, n (%) | Hospice                 | 5 (0.3)      |
|                                                                  |                           | Other                   | 112 (7.5)    |
|                                                                  |                           | Transfer                | 28 (1.9)     |
|                                                                  |                           | Government              | 100 (6.7)    |
|                                                                  |                           | Medicaid                | 116 (7.7)    |
|                                                                  | INSURANCE, n (%)          | Medicare                | 331 (22.1)   |
|                                                                  |                           | Private                 | 894 (59.6)   |
|                                                                  |                           | Self Pay                | 59 (3.9)     |



**Figure 3:** Data transformation of EHRs from (a) raw discharge notes (with extracted entities) to (b) constructed knowledge graph (through NER). Here, legend indicates medical entity and color mapping and only a subset of symptoms and medical history is presented in the figure for clarity. The patient’s family medical history is described as “Both parents died from CVA’s.” NER extracts “both parents” as Family Member, and “CVA” as Medical History. A relationship of “HAS\_FAMILY\_MEMBER” is established between the Patient node and Family Member node (both parents), and “HAS\_MEDICAL\_HISTORY” is established between the Family Member node and the Medical History node (CVA).



**Figure 1** shows the Reasoning RAG agentic workflow, where we provide the construction technical details of the agents in Supplementary Methods. The key stages in the workflow include *retrieval* with the Retrieval Agent and the KG Query Generation Agent, *reasoning* with the Abstraction Agent and the Checker Agent, and *generation* with the Rewrite Agent and the Summarization Agent. There are three initial inputs, including the AIPatient KG, a natural language query from the medical investigator, and the Conversation History between medical investigator and the SP. The first is only provided to the agents in the Retrieval Stage, and the latter two are universal inputs to all agents across the three stages.

For example, **Figure 4** shows a medical investigation that begins with a question input to the AIPatient system, “How long have you experienced soreness in your chest?” This question initiates a sequence of specialized agents that supports the *Reasoning RAG* process.

The Retrieval Stage begins with the Retrieval Agent that selects a subset of relevant nodes and edges of the AIPatient KG, based on the natural language query. In parallel, the Abstraction Agent [32] in the Reasoning Stage simplifies the specific or detailed user natural language query and paraphrases them into more generalized, high-level questions. Taking the inputs from Retrieval Agent and Abstraction Agent, the KG Query Generation Agent constructs a Cypher query [32], which is a declarative language particularly designed for Neo4j databases that can efficiently query and manipulate graph data through nodes, relationships, and properties. KG Query in Cypher is preferred over direct document RAG approaches. The queries can navigate complex graph structures, extracting rich and connected insights that are critical in understanding multidimensional relationships between medical entities. For example, symptoms and diseases are harder to capture with linear document-based retrieval methods. The final output of this stage is the Cypher query-retrieved information, such as symptom or disease intensity.

Next, the Checker Agent in the Reasoning Stage makes a decision of whether the retrieved information and the medical investigation inquiry aligns – if the Checker Agent approves the results (outputs “Yes”), the workflow proceeds to the Generation stage; if not (outputs “No”), the Checker Agent rephrases the natural language query based on the Conversation History and prompts the KG Query Generation Agent to regenerate the Cypher query. The checking process repeats for 3 times, and if no satisfactory information is retrieved after 3 iterations, the Generation Stage is skipped and the simulated patient outputs “I don’t know”.

The Generation Stage focuses on the alignment of medical information to realistic and diverse patient behaviors in responses. The Rewrite Agent aims to translate the technical KG query results into a more accessible and understandable natural language format, taking into account the specific personality traits assigned to the patient. It simulates the patient’s personality based on the Big Five personality traits model through prompting [33]. For single-round conversation, this output is passed back to the medical investigator. For multi-round conversation, the Summarization Agent then integrates the rewritten results and conversation and accordingly updates the conversation history.

## Evaluation and data labeling

Synthesizing the effectiveness and trustworthiness metrics in these recent research and applying them to the evaluation of AIPatient system, we focus on five evaluation dimensions presented in **Table 2**.

| <b>Table 2.</b> Evaluation framework |                              |                      |                                                 |
|--------------------------------------|------------------------------|----------------------|-------------------------------------------------|
| <b>Performance aspect</b>            | <b>Evaluation dimension</b>  | <b>Evaluation by</b> | <b>Metrics</b>                                  |
| Effectiveness                        | Knowledgebase validity (NER) | Medical doctors      | F1                                              |
|                                      | QA accuracy (conversation)   | Researchers          | Accuracy                                        |
|                                      | Readability                  | Algorithm            | Flesch Reading Ease, Flesch-Kincaid Grade Level |
| Trustworthiness                      | Robustness (system)          | Researchers          | Accuracy, ANOVA                                 |
|                                      | Stability (personality)      | Researchers          | Accuracy, ANOVA                                 |

We created two evaluation datasets to assess the five performance aspects. First, to evaluate LLM’s performance for generating the system’s information input, i.e., Knowledgebase Validity (NER), we created a gold-standard labeled data with expert medical practitioners using the Doccano annotation interface (detailed in **Supplementary Figure 2**). 100 cases were selected and six physicians independently labeled the medical entities of interest, including symptom, medical history, vitals, allergies, family history, and social history. Each case was labeled by two physicians and we perform the Inter-coder Reliability Check by calculating span-level F1 scores, which stands at 0.79. The F1 score provides a balance between precision and recall, and evaluating it at the span level takes into account the exact start and end positions of entities. This makes span-level evaluation more rigorous than simple token-based assessments, as it ensures that both the correct entity type and its precise boundaries within the text are accurately identified. Finally, a separate MD that did not participate in the data labeling process reconciled the differences between the labeling. We also used this NER task to benchmark and select LLM for the downstream tasks.

For the rest of the performance aspects, including QA Accuracy, Readability, Robustness and Stability, we developed a medical QA conversation dataset consisting of 524 questions. We first sampled 56 patient records from MIMIC-III and two Natural Language Processing (NLP) researchers reviewed their discharge summaries. Based on this review, we formulated the questions to focus on medical entities within the records. An example for a symptom-related question is: “what is the duration of the symptom ‘chest pain’?” A subset of the questions are focused on complete set retrieval, e.g., “What are all the symptoms that the patient with SUBJECT\_ID = 22265 and HADM\_ID = 147802 have?” These questions test how accurate, readable, robust, and stable the AIPatient system is, with a variety of input formats,, including generic and iterative paraphrasing on the same set of questions.

## LLM selection and knowledgebase validity (NER) evaluation

Data use agreement of MIMIC-III strictly prohibits the use of Protected Health Information (PHI) data with online services such as GPT. To conform with data use requirements, we use the Azure OpenAI service for GPT-family LLM and Amazon Bedrock for Claude-family LLM, both not sharing data for model training [34].

To find the best performing model for the down-stream tasks, we use the gold-standard labeled dataset to evaluate six models' performance on the knowledge validity (NER) tasks. The models include three Claude models (Claude-3.5 Sonnet, Claude-3 Haiku, and Claude-3 Sonnet) [35] and three GPT models (GPT-4 Turbo, GPT-4o, and GPT-3.5 Turbo) [36–38].

To perform NER tasks, we designed prompts for extracting each type of medical entity detailed in Supplementary Methods. We calculated general and medical entity category-specific True Positive Rate, False Positive Rate, Precision, Recall and F1 as follows:

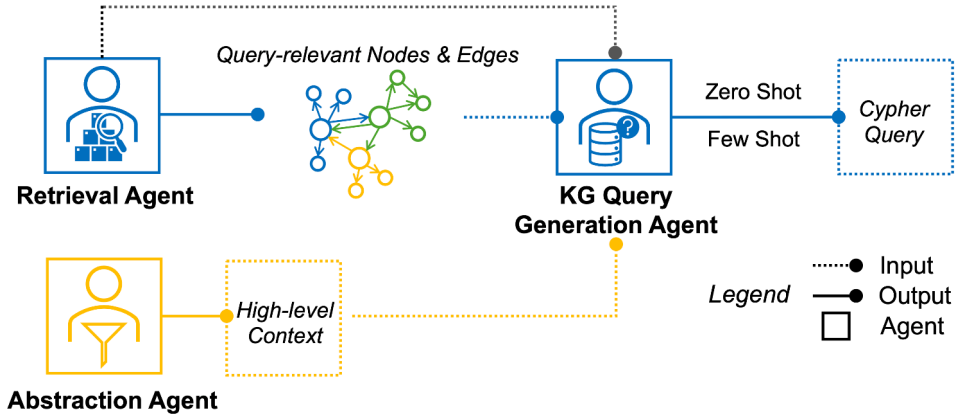
- **True Positive Rate:**  $TP/(TP + FN)$
- **False Positive Rate:**  $FP/(TN + FP)$
- **Precision:**  $TP/(TP + FP)$
- **Recall:**  $TP/(TP + FN)$
- **F1:**  $(2 * Precision * Recall) / (Precision + Recall)$

Here, we specify the in-context definitions of TP, FP, TN, and FN:

- TP: Number of instances where both LLM and gold-standard data agree a medical entity is present.
- FP: Number of instances where LLM indicates a medical entity is present, but gold-standard data shows no medical entity.
- TN: Number of instances where both LLM and gold-standard data agree no entity is present.
- FN: Number of instances where LLM indicates no medical entity, but gold-standard data shows a medical entity is present.

## QA accuracy evaluation through ablation studies

**Figure 5** highlights a section of the AIPatient system on which we conduct the QA conversation accuracy evaluation. We conduct ablation studies for the KG Query Generation Agent in different scenarios (1) on its own, (2) in combination with the Retrieval Agent, the Abstraction Agent, or both, and (3) using zero-shot or few-shot strategy. Using an ablation study framework, there are a total of eight setups. For all setups, two researchers independently compare the retrieved results against the correct answers, and provide binary ratings of correct or incorrect. To ensure the reliability and consistency of the evaluation process, an intercoder reliability check is performed with Cohen's Kappa of 0.92 [39]. To resolve discrepancies in data labeling, we then include a third annotator, who is an experienced researcher in medical AI, to judge the annotation difference and decide the correct label. We use the accuracy rate to evaluate the model's ability to precisely and reliably answer questions based on EHRs. We define accuracy rate as the number of questions LLM answered correctly over the total number of questions.



**Figure 5.** Focused Section of Cypher Query Evaluation

## Readability evaluation

The primary goal of AIPatient is to simulate realistic patient interactions in medical training and practice. For the simulations to be effective, the output must be easily understandable by users of varying expertise, including students, trainees, and seasoned medical professionals. We focus on the output of the Rewrite Agent, and employ two metrics based on Average Sentence Length (ASL, the number of words per sentence) and Average Syllables per Word (ASW) to assess the readability of AIPatient and determine the ease of understanding for users.

The first metric is the Flesch Reading Ease [40], where Higher scores indicate easier readability, with a score of 90-100 suggesting text easily understood by an average 11-year-old, and scores of 30-50 indicating college-level material. Here,

- **Flesch Reading Ease** =  $206.835 - (1.015 \times \text{ASL}) - (84.6 \times \text{ASW})$ .

The second metric is the Flesch-Kincaid Grade Level [41], which estimates the U.S. school grade level required to understand the text. Lower grade levels correspond to simpler text, while higher grade levels reflect more complex material. Here,

- **Flesch-Kincaid Grade Level** =  $(0.39 \times \text{ASL}) + (11.8 \times \text{ASW}) - 15.59$ .

For each question in the QA set, we prompt the Rewrite Agent to write the answer in the context of the question. This step mimics the Rewrite process in the AIPatient framework, where the Rewrite Agent will formulate natural language responses to user's questions based on the Cypher query results. We then calculate the Flesch Reading Ease and Flesch-Kincaid Grade Level for each of the rewritten responses.

## Robustness (system) evaluation

Robustness is a critical component of the trustworthiness of AIPatient, reflecting the system's reliability and consistency across varied inputs. To evaluate the robustness of AIPatient, we employ a methodology centered around the resilience of the system to changes in the phrasing of questions and assess the system robustness.

Each question in the QA conversation set is paraphrased three times to create a diverse pool of queries that maintain the original intent but differ in structure and vocabulary.

This paraphrasing mimics the real-world scenario where different users may phrase the same clinical question in various ways.

To answer each paraphrased question, we invoke the Reasoning RAG framework and extract the required information from the knowledge graph. Finally, the robustness of AIPatient is assessed by comparing the accuracy of the responses generated from the paraphrased questions against the responses from the original questions (two sample t-test and ANOVA).

## Stability (personality) evaluation

The stability of personality in AIPatient is another vital aspect of its trustworthiness, particularly concerning the system's ability to integrate distinct personality traits into simulated patient interactions without compromising the accuracy or integrity of medical information. This evaluation focuses on ensuring that personality variations do not distort or omit essential clinical data, thereby maintaining consistency in the information presented across different simulated personalities.

Utilizing the Big Five personality framework, we generate 32 distinct personality types by combining different levels of the five major traits. These personality profiles are then integrated into the Rewrite Agent of AIPatient. The Rewrite Agent's task is to adjust the responses generated by the system to reflect the specific personality traits of the simulated patient while answering questions from the QA set. This involves altering the tone, style, and certain expressions in the answer to align with the given personality, without altering the core medical information.

Finally, researchers manually review the outputs to ensure that all personality-infused rewrites maintain the core information from the original answers. The primary goal is twofold:

1. To confirm that the inclusion of personality traits in the rewrites does not lead to a loss of essential information compared to rewrites without personality traits.
2. To check that there is no statistically significant difference in the accuracy of information across different personality groups.

To quantitatively assess these goals, we calculate the personality-induced data loss proportion, and employ ANOVA to test for statistical differences in the quality of information retention across the different personality profiles.

## Statistical analysis

In Robustness and Stability evaluation, we use ANOVA to statistically determine if there are significant differences in the system's performance across various input paraphrases and personality-infused responses. Two proportion t-tests are conducted to compare the accuracy of the system's responses between the original questions and their paraphrased counterparts, and between responses generated with and without personality traits. A two-sided  $P \leq 0.05$  was considered statistically significant. Statistical analyses were carried out using the statistical Python package in scipy (Scipy.org). Python version 3.9.16 (Python Software Foundation) was used to carry out this work.

# Results

## Knowledge graph statistics

The Retrieval Agent directly queries from the AIPatient KG and extracts entities in discharge notes in MIMIC-III. **Table 3** presents the statistics of entities in the AIPatient KG: there are 1,500 patient-admission records, with a total of 15,441 nodes and 26,882 edges.

| <b>Table 3</b> AIPatient KG Nodes and Edges Statistics |              |                      |               |
|--------------------------------------------------------|--------------|----------------------|---------------|
| <b>Nodes</b>                                           | <b>Count</b> | <b>Relationships</b> | <b>Counts</b> |
| Patient                                                | 1495         |                      |               |
| Admission                                              | 1500         | HAS_ADMISSION        | 1500          |
| Symptom                                                | 3955         | HAS_SYMPTOM          | 7000          |
|                                                        |              | HAS_NOSYMPTOM        | 2971          |
| Duration                                               | 683          | HAS_DURATION         | 1544          |
| Intensity                                              | 206          | HAS_FREQUENCY        | 383           |
| Frequency                                              | 190          | HAS_INTENSITY        | 408           |
| History                                                | 2485         | HAS_MEDICAL_HISTORY  | 3865          |
| Vital                                                  | 717          | HAS_VITAL            | 2425          |
| Allergy                                                | 227          | HAS_ALLERGY          | 596           |
| Social History                                         | 3287         | HAS_SOCIAL_HISTORY   | 4497          |
| Family Member                                          | 143          | HAS_FAMILY_MEMBER    | 885           |
| Family Medical History                                 | 553          | HAS_MEDICAL_HISTORY  | 808           |

## LLM selection and knowledgebase validity (NER)

For each of the medical entity categories, we experimented with all six LLMs and evaluated their NER-based knowledgebase validity. As **Table 4** shows, we observe superior knowledgebase validity for the GPT-4-Turbo model, with the highest overall F1 of 0.89. Results for TPR, FPR, Precision and Recall are presented in **Supplementary Table 1-4**. For the Claude family models, Claude-3.5-sonnet shows the best performance of 0.74 overall F1. We note that the GPT-family models specifically excelled in extracting Allergies, where the older version of Claude models suffered (Claude-3-haiku and Claude-3-sonnet). Based on these results, we use the GPT-4 Turbo model to construct the final version of AIPatient KG.

| <b>Table 4</b> NER F1 by Entity Categories   |                       |                        |                          |                      |               |                    |
|----------------------------------------------|-----------------------|------------------------|--------------------------|----------------------|---------------|--------------------|
| <b>Entity Categories</b>                     | <b>claude-3 haiku</b> | <b>claude-3 sonnet</b> | <b>claude-3.5 sonnet</b> | <b>gpt-3.5 turbo</b> | <b>gpt-4o</b> | <b>gpt-4 turbo</b> |
| Symptom Group <sup>1</sup>                   | 0.69                  | 0.70                   | 0.78                     | 0.72                 | 0.75          | <b>0.90</b>        |
| Medical History                              | 0.69                  | 0.69                   | 0.89                     | 0.87                 | 0.96          | <b>0.98</b>        |
| Allergies                                    | 0.68                  | 0.69                   | 0.70                     | 0.69                 | 0.74          | <b>0.87</b>        |
| Family and Social History Group <sup>2</sup> | 0.71                  | 0.71                   | 0.75                     | 0.71                 | 0.74          | <b>0.91</b>        |
| Total                                        | 0.69                  | 0.70                   | 0.74                     | 0.71                 | 0.75          | <b>0.89</b>        |

<sup>1</sup>Symptom Group includes symptoms (both symptoms and denied symptoms), duration, intensity and frequency.

<sup>2</sup>Family and Social History Group includes Family History (family members and their medical history) and Social History

## QA accuracy ablation studies

In **Table 5**, we highlight ablation studies results across Symptom, Medical History, Family and Social History. Results of additional medical categories (Admission, Patient, Allergy, Vitals) are presented in **Supplementary Table 5**, where all LLM achieved a high accuracy rate regardless of the setup chosen. The setup with all agents and few-shot learning achieves the highest accuracy in most categories, with 94.15% overall accuracy. The baseline without the AIPatient KG and Reasoning RAG performs worse, particularly in Family and Social History, where accuracy drops to 13.33%.

**Table 5** Ablation Studies Result by Medical Category (Differentiating Set)

| Few Shot                                                      | Retrieval Agent | Abstraction Agent | Overall       | Symptom Group | Medical History | Family and Social History |
|---------------------------------------------------------------|-----------------|-------------------|---------------|---------------|-----------------|---------------------------|
| ✓                                                             | ✓               | ✓                 | <b>94.15%</b> | <b>91.20%</b> | <b>87.10%</b>   | <b>85.56%</b>             |
| ✓                                                             | ✓               |                   | 92.60%        | 89.68%        | 83.87%          | 78.89%                    |
| ✓                                                             |                 | ✓                 | 93.80%        | 90.48%        | 83.87%          | <b>85.56%</b>             |
| ✓                                                             |                 |                   | 92.94%        | 90.48%        | 69.35%          | 82.22%                    |
|                                                               | ✓               | ✓                 | 81.41%        | 85.71%        | 25.81%          | 60.00%                    |
|                                                               | ✓               |                   | 81.93%        | 84.92%        | 27.42%          | 58.89%                    |
|                                                               |                 | ✓                 | 83.13%        | 87.20%        | 30.65%          | 64.44%                    |
| Only with <i>KG Query Generation Agent</i>                    |                 |                   | 82.62%        | 88.80%        | 25.81%          | 60.00%                    |
| Without <i>Reasoning RAG</i> &<br>Without <i>AIPatient KG</i> |                 |                   | 68.94%        | 64.29%        | 53.45%          | 13.33%                    |

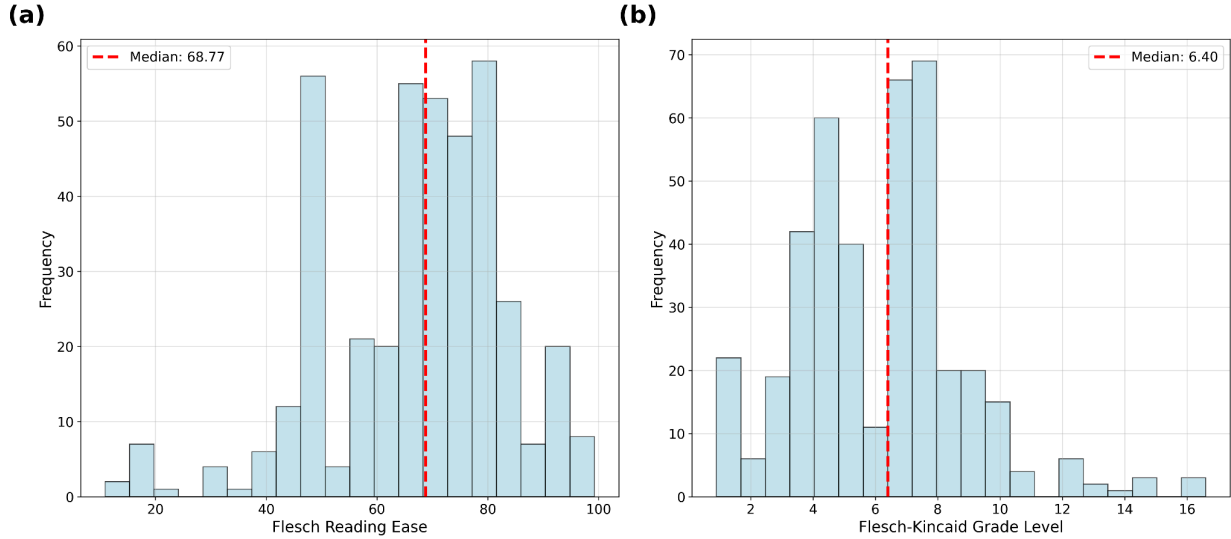
\* Highest accuracy in each category is in bold.

\*\* All results are with *AIPatientKG* as the input, unless specified otherwise.

\*\*\* This differentiating set excludes admission, patients, allergies and vitals, for which we observe 100% or close to 100% accuracy regardless of setup.

## Readability

**Figure 6(a)** shows that the median Flesch Reading Ease scores range from 10.91 to 99.23 (median 68.77). **Figure 6(b)** indicates that the Flesch-Kincaid Grade Level peaks at 6th grade level (median grade level 6.4).



**Figure 6:** Distribution of the Flesch Reading Ease and the Flesch-Kincaid Grade Level.

## Robustness (System)

The results of the ANOVA (**Table 6**) and the two-sample t test (**Supplementary Table 6**) reveal that for the *Overall* dataset, the QA conversation paraphrase did not significantly affect response accuracy, with an F-value of 0.6126 and a p-value of 0.5420. This indicates no significant differences between the three paraphrase groups and the original questions, suggesting that the system is robust to changes in question phrasing overall.

For specific sub-categories, in the *Medical History* category, we observe a significant difference with an F-value of 5.3038 and a p-value of 0.00589, indicating that the system's accuracy in this category is more sensitive to the phrasing of questions. This may be due to the complexity or specificity of medical history-related queries. For the *Symptom* and *Family and Social History* categories, no significant differences were found, further indicating the system's resilience in handling varied input phrasings in these areas.

| <b>Table 6</b> ANOVA Results for Robustness Evaluation on Three Paraphrase Sets |                       |                          |          |                |
|---------------------------------------------------------------------------------|-----------------------|--------------------------|----------|----------------|
| <b>Medical Category</b>                                                         | <b>Sum of Squares</b> | <b>Degree of Freedom</b> | <b>F</b> | <b>P-Value</b> |
| Overall                                                                         |                       |                          |          |                |
| Paraphrase Group                                                                | 0.0818                | 3                        | 0.6126   | 0.5420         |
| Residual                                                                        | 102.9382              | 1541                     |          |                |
| Symptom                                                                         |                       |                          |          |                |
| Paraphrase Group                                                                | 0.0821                | 3                        | 0.54     | 0.5835         |
| Residual                                                                        | 16.4383               | 216                      |          |                |
| Medical History                                                                 |                       |                          |          |                |
| Paraphrase Group                                                                | 1.7160                | 3                        | 5.3038   | 0.00589        |
| Residual                                                                        | 25.7222               | 1596                     |          |                |
| Family and Social History                                                       |                       |                          |          |                |
| Paraphrase Group                                                                | 0.0896                | 3                        | 0.2536   | 0.7761         |
| Residual                                                                        | 41.7094               | 235                      |          |                |

## Stability (Personality)

The 32 personality groups have a median personality-induced data loss proportion of 2%, and range of 0% to 5.88%, indicating consistently low data loss proportion across all personality groups, with only a small range of variability.

As shown in **Table 7**, for the *Overall* dataset, the paraphrase group did not significantly affect the system’s performance, with an F-value of 0.7820 and a p-value of 0.7990, indicating no substantial differences across personality groups. Similarly, no significant differences were observed in the *Symptom* ( $F = 1.1104$ ,  $p = 0.3230$ ) and *Family and Social History* ( $F = 0.6774$ ,  $p = 0.9024$ ) categories. For the *Medical History* category, no statistical test was conducted due to the lack of variation, as all personality groups achieved 100% accuracy. These results suggest that the AIPatient system maintains its stability across different simulated personalities, ensuring the integrity and consistency of the medical information presented, regardless of personality variations.

**Table 7: ANOVA Results for Stability Evaluation on 32 Personality Groups**

| Medical Category          | Sum of Squares | Degree of Freedom | F      | P-Value |
|---------------------------|----------------|-------------------|--------|---------|
| Overall                   |                |                   |        |         |
| Paraphrase Group          | 0.3529         | 31                | 0.7820 | 0.7990  |
| Residual                  | 23.2941        | 1600              |        |         |
| Symptom                   |                |                   |        |         |
| Paraphrase Group          | 1.4023         | 31                | 1.1104 | 0.3230  |
| Residual                  | 9.1250         | 224               |        |         |
| Family and Social History |                |                   |        |         |
| Paraphrase Group          | 0.9023         | 31                | 0.6774 | 0.9024  |
| Residual                  | 9.6250         | 224               |        |         |

\* For the Medical History category, all personality groups have 0% data loss proportion; no statistical test is conducted due to lack of variation.

## Discussion

We developed an EHRs and LLM powered patient simulation system, the AIPatient system, that leverages the *Reasoning RAG* agentic workflow and the MIMIC-III-based AIPatient KG to support medical investigation in an effective and trustworthy manner. Overall, this system facilitates complex reasoning through LLM agent interactions and accurate patient information processing with NER, potentially allowing for dynamic and human-like responses to medical questions across a range of simulated patient personalities. The *Reasoning RAG* framework extends the current RAG strategy and optimizes system performance through stepwise and interactive agentic instructions with few-shot examples, feedback loops, and conversation history updates. The construction of the AIPatient KG prototypes a large-scale NER-based EHR processing method, which also enables LLM selection for downstream tasks. From the perspective of system assessment strategies, we further contribute comprehensive evaluation guidelines for medical NLP systems, especially useful for EHRs and LLM based multi-agent systems.

## Knowledge graph enables structured medical information retrieval from EHRs

The primary objective of NER is to extract structured entities from unstructured clinical notes or patient records [42]. By extracting medical entities and their interrelationships from this dataset, the AIPatient KG serves as a reliable backbone of the AIPatient system. Using a knowledge graph to preprocess and organize EHRs data offers several advantages over a traditional free text or tabular format.

First, compared to the unstructured clinical notes, knowledge graphs represent a structured representation of entities and relationships. This structured approach facilitates clearer organization and retrieval of information, enabling more efficient data analysis and reasoning. Here, NERs enhance knowledge graphs by correctly key entities such as diseases and medications from unstructured clinical notes, enabling the systematic and correct organization of these entities into a structured format with a F1 score of 0.89.

Second, knowledge graphs allow for a more intuitive and interconnected view of data. In clinical contexts, this means easily visualizing how patients, symptoms, diseases, treatments, and outcomes are related. Users can perform more complex queries that traverse these relationships, such as finding all patients with a specific frequency who were treated with a particular medication and had certain outcomes. This relationship is less apparent in free text or tabular data.

Third, unlike tabular data, knowledge graphs can evolve their schema without major rework. New types of entities and relationships can be added as medical knowledge and needs evolve. This is desirable specifically for the AIPatient project, as we are continuously exploring the MIMIC-III dataset for meaningful data sources and relationships.

Lastly, knowledge graphs offer concise and efficient organization and representation of information. When integrated with large language models, they mitigate catastrophic forgetting and reduce the high tokenization costs associated with processing vast amounts of information.

## LLM Agentic workflow with abstractive and self-reflective reasoning enhances medical QA accuracy and realism

Agentic workflow involves more interactive, iterative, and multi-step task executions rather than the traditional one-off responses to user prompts [43]. This method leverages the collective capabilities of AI agents, prompt engineering, and generative AI networks to create a dynamic environment where AI agents are not just responders but active participants in problem-solving processes [43]. The core elements of agentic workflows include self-reflection (e.g., Checker Agent), tool utilization (e.g., Abstraction and Rewrite Agents), and multi-agent collaboration, each adding layers of depth and functionality that enhance the overall effectiveness and intelligence of AI systems [44]. Our *Reasoning RAG* agentic framework improves accuracy and minimizes hallucination risk.

Reasoning RAG relies on abstraction and iterative checking for enhanced reasoning. The reasoning process in the RAG framework involves abstraction, where complex medical data is distilled into manageable pieces for LLMs, improving performance. Complementing this,

the checking step introduces iterative self-reflection, enabling AI agents to refine their responses by cross-checking data. The dynamic interplay between the checker agent and the KG query generation agent further boosts the system’s reliability and relevance in simulated medical environments, yielding an accuracy rate of over 94.15%.

By integrating few-shot learning and personality aspects, AIPatient delivers human-like, accessible, and context-aware generation while maintaining high system robustness and personality stability in responses to medical investigations. Notably, from **Table 5**, we observe a systematic improvement in accuracy with the few-shot strategy. On average, the accuracy rate increased by 11.1% after incorporating few-shot examples into the prompt. We also note that on average, accuracy is the lowest for answering social history questions. This is likely due to the result of the variety of social history representations (i.e., “he never smoked,” “3 packs a day,” or “quit tobacco.”). High readability is another important aspect for facilitating effective learning and interaction, ensuring that medical information is conveyed clearly without overwhelming the user with overly technical language. These results highlight that the AIPatient system successfully balances the need for medical accuracy with the necessity for clear communication, making it potentially useful for both medical student training and medical system integrations.

## Limitations and future work

While the AIPatient system currently benefits from the *Reasoning RAG* framework and a multi-agent workflow, further enhancements could improve user experience. In particular, we aim to develop and integrate an automatic Evaluator Agent to assist with scoring and providing feedback during medical investigations. This Evaluator Agent could reference medical textbooks and clinical guidelines, providing evidence-based assessments to ensure accuracy and alignment with current medical standards [6]. By incorporating this feature, the system could offer detailed, standardized feedback that medical students and professionals could rely on for learning and improvement.

In addition, our present implementation of the AIPatient KG relies heavily on discharge notes from the MIMIC-III dataset, which primarily covers critical care cases. This presents a limitation, as it lacks representation of a broader set of scenarios, including those encountered in outpatient, inpatient, and long-term care settings. Expanding the database to include EHRs from these additional domains would enhance the diversity of patient cases, offering medical students exposure to a wider range of clinical conditions. At the same time, the current dataset only covers a homogenous population from Boston, Massachusetts, which limits the generalizability of this study’s results when applied to other datasets. In future work, we plan to enrich the KG by incorporating more medical entities and relationships, such as patient histories of medication use, negative reports on allergies, prescriptions, and detailed medical histories. We will also expand the current patient population in MIMIC-III to a broader population to enhance system inclusiveness. These expansions would provide a more comprehensive simulation experience, aligning better with the complex reality of patient care.

Similarly, the *Reasoning RAG* can also benefit from additional expansions, including multimodal integration, processing speed optimization, and data privacy and security. Currently, our system operates using textual data. However, the advent of multimodal large language

models (MLLMs) that can process a variety of data types opens new possibilities. Future iterations of AIPatient could integrate medical images such as ECGs, X-rays, MRIs, and CT scans, allowing for a richer and more holistic patient simulation experience. By incorporating these modalities, the system could offer more realistic and comprehensive medical investigations, where both clinical notes and imaging data are used in diagnosis and treatment planning.

Another challenge faced by the current system is its relatively slow processing speed, caused by the need for multiple agents to operate sequentially via API calls. This results in longer wait times for completing simulations. To address this, we are exploring several optimizations. First, we plan to implement local finetuned LLM or Small Language Models (SLM) to reduce reliance on external API calls, which will enable faster, more efficient interactions. This approach will not only improve the speed of the system but also enhance the overall user experience by enabling more seamless, real-time responses. Implementing a locally-hosted model will also address concerns related to data privacy and security. By maintaining end-to-end control of the model, the system can ensure that sensitive patient information remains secure, avoiding the need to transmit data over external networks. This is especially important in clinical settings, where patient confidentiality is paramount.

While our current evaluation framework is comprehensive, it can further benefit from multi-round evaluation and risk assessments. At present, the evaluation process for AIPatient is limited to single-round interactions. However, many real-world medical investigations require iterative questioning and decision-making processes. To address this, future versions of the system could include multi-round evaluations, designed with input from professionals across various medical disciplines to better reflect the nuances of their respective fields. An additional area for future research could be the assessment of risks within the system. While current evaluation methods focus on the system's application risk in accurately role-playing as patients, we must also ensure that simulated patients reflect appropriate socio-cultural contexts and realistic case scenarios. Addressing these issues could minimize the risk of significant errors and enhance the system's safety and reliability in real-world applications.

Finally, future work is needed to understand the comfort and concerns of trainees, physicians, and patients regarding the implementation of such generative AI systems in clinical education and practice. Although the AIPatient system offers significant enhancements to medical training, the integration of AI in sensitive healthcare contexts brings potential ethical, psychological, and professional considerations. For instance, medical trainees and practitioners may have concerns about the system's impact on their learning process, the accuracy and reliability of AI-provided feedback, and the potential replacement of traditional training methods. Similarly, patients may have reservations about AI's involvement in simulated clinical scenarios and how it might influence care quality. To address these aspects, future research should explore user experiences and gather feedback from trainees, clinicians, and patients, ensuring that the system evolves in a way that aligns with the expectations and comfort levels of all stakeholders involved.

## Data Availability

MIMIC-III data were accessed through PhysioNet [45]. A data usage agreement was required to obtain the data. LLM were used in compliance with PhysioNet standard, including using the GPT models on Azure and the Claude models on Amazon Bedrock for data privacy.

## Code Availability

The code and a demo are publicly available on GitHub:

<https://github.com/huiziy/AIPatient2024>. An updated full code base will be released.

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## Author Contributions

H.Y.: conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing—original draft, writing—review & editing. J.Z. and L.Y.L.: data curation, formal analysis, investigation, methodology. S.C., J.G., A.S., X.L., W.H., M.J.: data curation, investigation, writing—review & editing. G.C., Y.Z., Z.L., T.G., M-L.C., and Z.A.: data curation, investigation, methodology. Y.Z., T.L.A., X.M.: conceptualization, investigation, methodology, writing—review & editing. D.S.B.: conceptualization, data curation, formal analysis, investigation, methodology, supervision, writing—original draft, writing—review & editing. L.L.: conceptualization, supervision, writing—review & editing. L.F.: funding acquisition, conceptualization, data curation, formal analysis, investigation, methodology, supervision, writing—original draft, writing—review & editing.

## Competing interests

H.Y.: Employment as a data scientist and stockholder in Amazon.com, Inc., unrelated to this work. D.S.B.: Associate Editor of Radiation Oncology, HemOnc.org (no financial compensation, unrelated to this work); funding from American Association for Cancer Research (unrelated to this work). All other authors declare they have no competing interests.

# Supplementary Information

## Supplementary Methods

### LLM Parameter Details

#### **LLM used for AIPatient Performance Evaluation Tasks**

1. LLM Selection and knowledgebase Validity (NER)
  - a. Claude-3 haiku
  - b. Claude-3 sonnet
  - c. Claude-3.5 sonnet
  - d. GPT-3.5 turbo
  - e. GPT-4o
  - f. GPT-4 turbo
2. QA Conversation Accuracy Ablation Study
  - a. GPT-4 turbo
3. Readability
  - a. GPT-4 turbo
4. Robustness (System)
  - a. GPT-4 turbo
5. Stability (Personality)
  - a. GPT-4 turbo

#### **LLM parameters**

1. max\_tokens\_to\_sample = 4096
2. temperature = 1

## Prompt Engineering Strategy

We enable these strategies when constructing the prompts for optimizing the performance of various agents within the system. These strategies support better interaction and more accurate retrieval, generation and simulation of patients.

1. **Role specification** [46]: Role specification is often used to set context for the task at hand. In the case of biomedical domains, defining the model's role as a biomedical researcher helps guide the LLMs to focus on relevant terms and relationships specific to that domain. Using predefined roles aligns with studies showing that role-specific context boosts performance in biomedical text mining and entity recognition.
2. **Few-shot** [47,48]: We include a few examples in the prompts, which allow the models to generalize effectively by drawing on prior knowledge. This approach significantly reduces the need for large volumes of annotated medical data while still maintaining high accuracy in tasks like entity recognition and relationship extraction.
3. **XML-style tag** [49,50]: To improve entity extraction and relationship mapping, we employ XML-style tagging in our prompts. This structured format ensures that the output is both readable and easily processed.

## NER Prompt Engineering Details

AlPatient KG is constructed from the medical entities (nodes) and relationships (edges) extracted from the unstructured clinical notes. Relationships illustrate how entities are interconnected, such as “patient has certain symptoms” and “disease has specific duration.” These extractions are based on an LLM powered NER approach, using GPT-4 model. In particular, we use prompts with the following characteristics:

*The specific prompt will be available later*

## AlPatient KG Schema

The full schema of the AlPatient KG is as follows. This schema is included in the prompt to allow agents to effectively query and retrieve relevant EHR data from the knowledge graph.

*The specific prompt will be available later*

## Retrieval Agent Details

- conversation history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.
- user query: the input query from the user. For example: "Can you tell me your symptoms?"
- schema: the general structure of the knowledge graph, including node properties and relationships, as specified in the supplementary methods of AIPatient KG Schema. For example: Patient node has properties: SUBJECT\_ID, GENDER, AGE, ETHNICITY, RELIGION, MARITAL\_STATUS; relationship: FamilyMember -> FamilyMedicalHistory: HAS\_MEDICAL\_HISTORY

*The specific prompt will be available later*

## Abstraction Agent Details

- user\_query: the input query from the user. For example: “Can you tell me your symptoms?”
- conversation\_history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.

*The specific prompt will be available later*

## KG Query Generation Agent Details

An example of cypher query generation prompt (with few-shot, abstraction agent, and retrieval agent input) is listed below:

- subject\_id: unique patient identifier
- hadm\_id: unique admission identifier
- nodes\_edges: output of the Retrieval Agent, contains relevant nodes and edges for constructing the cypher query. For example: {'Nodes': ['Allergy'], 'Relationships': ['HAS\_ALLERGY']}
- conversation\_history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.
- user\_query: the input query from the user. For example: "Can you tell me your symptoms?"
- schema: the general structure of the knowledge graph, including node properties and relationships, as specified in the supplementary methods of AIPatient KG Schema. For example: Patient node has properties: SUBJECT\_ID, GENDER, AGE, ETHNICITY, RELIGION, MARITAL\_STATUS; relationship: FamilyMember -> FamilyMedicalHistory: HAS\_MEDICAL\_HISTORY
- abstraction\_context: output of the Abstraction Agent, contains step-back context. For example, when the user ask about whether the patient has fever, the step back context is a list of all the symptoms the patient has.

*The specific prompt will be available later*

## Checker Agent Details

An example of Checker Agent prompt is provided below:

- user\_query: the input query from the user. For example: “Can you tell me your symptoms?”
- conversation\_history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.
- query\_result: extracted result from AIPatient KG based on the user\_query. For example, ['black and bloody stools', 'lightheadedness', 'shortness of breath'] for the user query: “what symptoms do you have?”

*The specific prompt will be available later*

## Rewrite Agent Details

This rewrite agent includes five broad dimensions: openness, conscientiousness, extraversion, agreeableness, and neuroticism. Each patient is assigned specific traits within these dimensions to create a realistic and relatable personality profile. For example, a patient with personality traits of “sensitive/nervous” in the neuroticism category might repeatedly ask for reassurance about their condition and treatment plan, and another with “friendly/compassionate” in the agreeableness category might be open to engaging in more detailed conversations and expressing gratitude for the care provided. An example of Rewrite Agent prompt is provided below:

- personality: the selected Big Five personality profile. It is a combination of five dimensions including: extraversion, agreeableness, openness, conscientiousness, and neuroticism. An example of one personality profile is [Practical, conventional, prefers routine, Impulsive, careless, disorganized, Quiet, reserved, withdrawn, Critical, uncooperative, suspicious, Calm, even-tempered, secure]
- user\_query: the input query from the user. For example: “Can you tell me your symptoms?”
- conversation\_history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.
- query\_result: extracted result from AIPatient KG based on the user\_query. For example, ['black and bloody stools', 'lightheadedness', 'shortness of breath'] for the user query: “what symptoms do you have?”

*The specific prompt will be available later*

## Summarization Agent Details

- conversation\_history: summarized conversation history between AIPatient and user. For example: Patient ID 23709 (Admission ID 182203) initially reported black and bloody stools. The patient has a complex medical history including depression, heart and brain issues, stomach troubles, and previous falls with broken bones. When asked about allergies, the patient disclosed an allergy to SSRI medications but expressed reluctance to discuss it further. The current context relates to the patient's allergy history.
- query\_result: extracted result from AIPatient KG based on the user\_query. For example, ['black and bloody stools', 'lightheadedness', 'shortness of breath'] for the user query: "what symptoms do you have?"
- patient\_response: natural language response from the simulated patient. For example: "Yes, I am over 50. But why does that matter right now? I prefer to stick to the routine questions we usually go through. And honestly, I don't see how my age is relevant to what we're discussing today. Can we move on to something more practical?"

*The specific prompt will be available later*

## Supplementary Tables

**Supplementary Table 1.** NER True Positive Rate (TPR) by Entity Categories

| Entity Categories                               | claude-3<br>haiku | claude-3<br>sonnet | claude-3.5<br>sonnet | gpt-3.5-<br>turbo | gpt-4o | gpt-4-<br>turbo |
|-------------------------------------------------|-------------------|--------------------|----------------------|-------------------|--------|-----------------|
| Symptom Group <sup>1</sup>                      | 39.83%            | 36.91%             | 41.64%               | 21.98%            | 67.59% | <b>83.85%</b>   |
| Medical History                                 | 18.58%            | 33.63%             | 62.89%               | 36.97%            | 59.72% | <b>79.48%</b>   |
| Allergies                                       | 4.35%             | 4.35%              | 89.86%               | 83.33%            | 95.65% | <b>100.0%</b>   |
| Family and Social History<br>Group <sup>2</sup> | 22.43%            | 31.73%             | 61.58%               | 32.04%            | 54.8%  | <b>87.92%</b>   |
| Total                                           | 31.04%            | 36.31%             | 53.79%               | 29.0%             | 63.81% | <b>84.21%</b>   |

<sup>1</sup>Symptom Group includes symptoms (both symptoms and denied symptoms), duration, intensity and frequency.

<sup>2</sup>Family and Social History Group includes Family History (family members and their medical history) and Social History

**Supplementary Table 2. NER False Positive Rate (FPR) by Entity Categories**

| Entity Categories                               | claude-3<br>haiku | claude-3<br>sonnet | claude-3.5<br>sonnet | gpt-3.5-<br>turbo | gpt-4o | gpt-4 |
|-------------------------------------------------|-------------------|--------------------|----------------------|-------------------|--------|-------|
| Symptom Group <sup>1</sup>                      | 42.06%            | 41.86%             | 38.49%               | 27.86%            | 30.19% | 9.92% |
| Medical History                                 | 33.39%            | 43.22%             | 21.17%               | 33.65%            | 30.29% | 3.02% |
| Allergies                                       | 0.0%              | 0.0%               | 19.05%               | 16.0%             | 8.7%   | 4.35% |
| Family and Social History<br>Group <sup>2</sup> | 29.84%            | 28.31%             | 26.55%               | 29.35%            | 28.17% | 4.04% |
| Total                                           | 37.49%            | 37.51%             | 28.59%               | 25.99%            | 27.96% | 6.47% |

<sup>1</sup>Symptom Group includes symptoms (both symptoms and denied symptoms), duration, intensity and frequency.

<sup>2</sup>Family and Social History Group includes Family History (family members and their medical history) and Social History

**Supplementary Table 3. NER Precision by Entity Categories**

| Entity Categories                               | claude-3<br>haiku | claude-3<br>sonnet | claude-3.5<br>sonnet | gpt-3.5-<br>turbo | gpt-4o | gpt-4-<br>turbo |
|-------------------------------------------------|-------------------|--------------------|----------------------|-------------------|--------|-----------------|
| Symptom Group <sup>1</sup>                      | 0.89              | 0.81               | 0.85                 | 0.86              | 0.81   | <b>0.98</b>     |
| Medical History                                 | <b>1.00</b>       | <b>1.00</b>        | 0.86                 | 0.92              | 0.95   | 0.96            |
| Allergies                                       | 0.78              | 0.82               | 0.81                 | 0.92              | 0.76   | <b>0.92</b>     |
| Family and Social History<br>Group <sup>2</sup> | 0.89              | 0.86               | 0.78                 | <b>0.90</b>       | 0.79   | 0.96            |
| Total                                           | 0.84              | 0.83               | 0.81                 | 0.91              | 0.79   | <b>0.95</b>     |

<sup>1</sup>Symptom Group includes symptoms (both symptoms and denied symptoms), duration, intensity and frequency.

<sup>2</sup>Family and Social History Group includes Family History (family members and their medical history) and Social History

**Supplementary Table 4. NER Recall by Entity Categories**

| Entity Categories                               | claude-3<br>haiku | claude-3<br>sonnet | claude-3.5<br>sonnet | gpt-3.5-<br>turbo | gpt-4o | gpt-4       |
|-------------------------------------------------|-------------------|--------------------|----------------------|-------------------|--------|-------------|
| Symptom Group <sup>1</sup>                      | 0.56              | 0.62               | 0.72                 | 0.62              | 0.69   | <b>0.83</b> |
| Medical History                                 | 0.53              | 0.53               | 0.92                 | 0.82              | 0.98   | <b>1.00</b> |
| Allergies                                       | 0.61              | 0.59               | 0.62                 | 0.55              | 0.72   | <b>0.82</b> |
| Family and Social History<br>Group <sup>2</sup> | 0.59              | 0.61               | 0.72                 | 0.59              | 0.69   | <b>0.86</b> |
| Total                                           | 0.59              | 0.60               | 0.67                 | 0.58              | 0.72   | <b>0.84</b> |

<sup>1</sup>Symptom Group includes symptoms (both symptoms and denied symptoms), duration, intensity and frequency.

<sup>2</sup>Family and Social History Group includes Family History (family members and their medical history) and Social History

**Supplementary Table 5.** System performance on the non-differentiating set of medical record categories

| Few Shot                                        | Retrieval Agent | Abstraction Agent | Admission      | Patient        | Allergy       | Vitals        |
|-------------------------------------------------|-----------------|-------------------|----------------|----------------|---------------|---------------|
| ✓                                               | ✓               | ✓                 | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
| ✓                                               | ✓               |                   | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
| ✓                                               |                 | ✓                 | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
| ✓                                               |                 |                   | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
|                                                 | ✓               | ✓                 | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
|                                                 | ✓               |                   | <b>100.00%</b> | <b>100.00%</b> | <b>96.67%</b> | <b>94.74%</b> |
|                                                 |                 | ✓                 | <b>100.00%</b> | <b>100.00%</b> | 90.00%        | <b>94.74%</b> |
| Only with KG Query Generation Agent             |                 |                   | <b>100.00%</b> | 98.28%         | <b>96.67%</b> | <b>94.74%</b> |
| Without Reasoning RAG &<br>Without AIPatient KG |                 |                   | 100.00%        | 87.50%         | 48.08%        | 90.00%        |

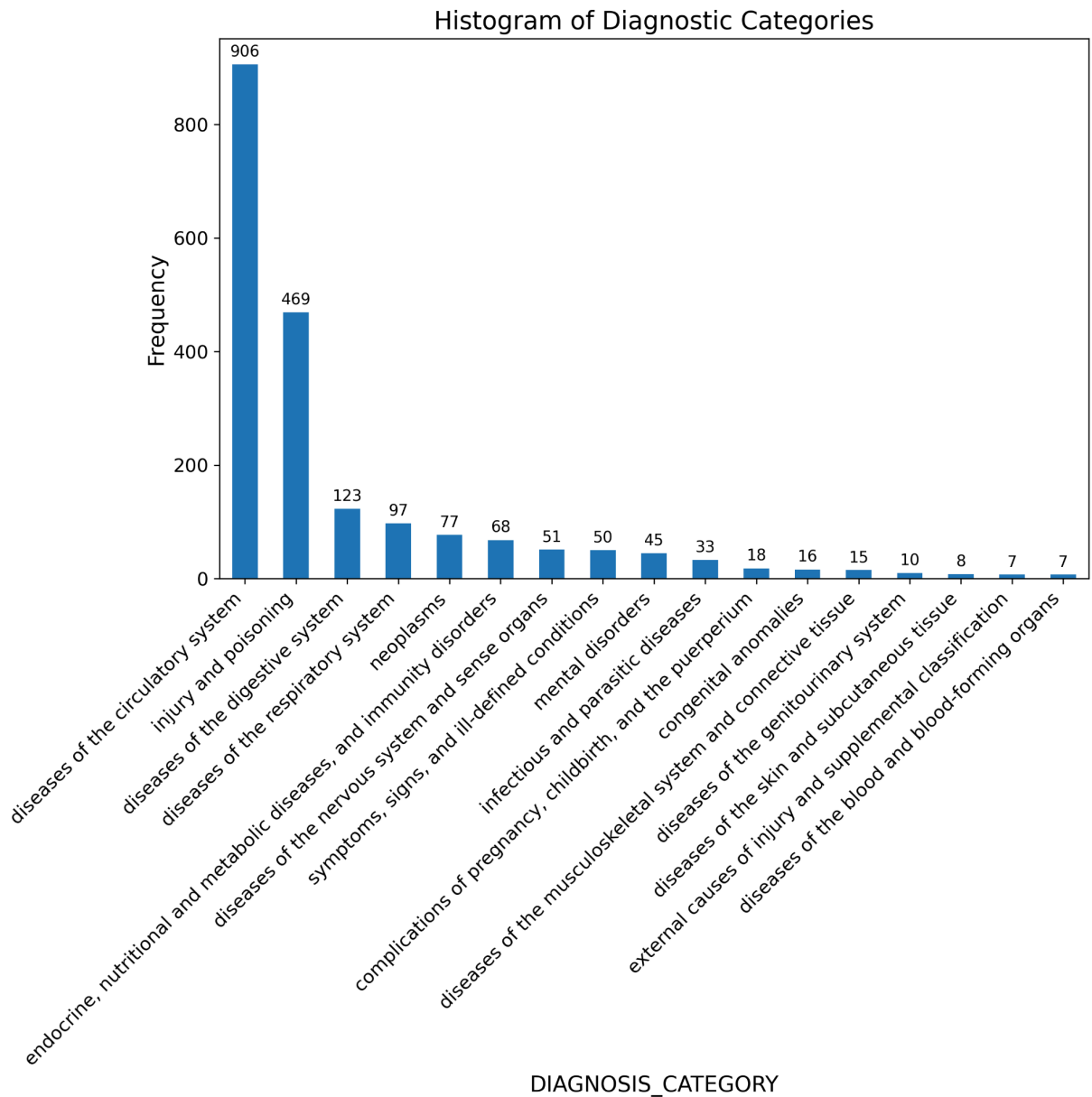
\* Highest accuracy in each category is in bold.

**Supplementary Table 6.** Two-Sample T-test of Accuracy Difference Between Paraphrase Question and Original Question

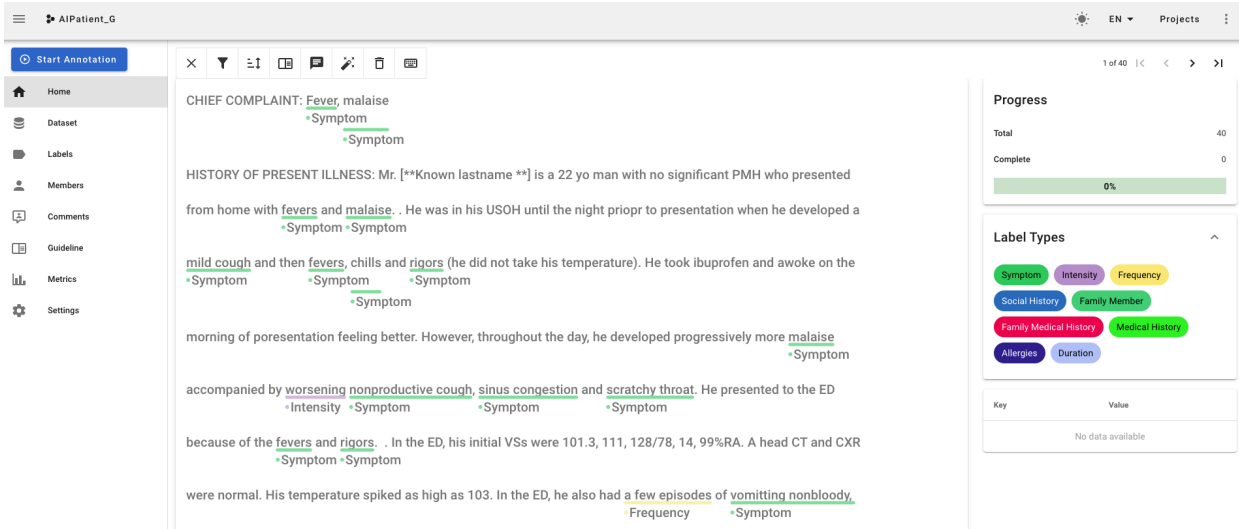
| Question Category         | Original Proportion | Paraphrase Proportion | Difference | Z=Statistics | P-Value |
|---------------------------|---------------------|-----------------------|------------|--------------|---------|
| Paraphrase 1              |                     |                       |            |              |         |
| Overall                   | 94.15%              | 93.79%                | 0.19%      | 0.13         | 0.90    |
| Symptom                   | 91.20%              | 93.15%                | 1.37%      | -0.29        | 0.77    |
| Medical History           | 87.10%              | 92.59%                | 3.70%      | -0.61        | 0.54    |
| Family and Social History | 85.56%              | 75.00%                | 10.00%     | 1.58         | 0.11    |
| Paraphrase 2              |                     |                       |            |              |         |
| Overall                   | 94.15%              | 92.04%                | 1.94%      | 1.22         | 0.22    |
| Symptom                   | 91.20%              | 93.15%                | 1.37%      | -0.29        | 0.77    |
| Medical History           | 87.10%              | 68.52%                | 18.52%     | 2.31         | 0.02    |
| Family and Social History | 85.56%              | 77.50%                | 7.50%      | 1.22         | 0.22    |
| Paraphrase 3              |                     |                       |            |              |         |
| Overall                   | 94.15%              | 92.61%                | 1.55%      | 0.99         | 0.32    |
| Symptom                   | 91.20%              | 89.04%                | 2.74%      | 0.53         | 0.60    |
| Medical History           | 87.10%              | 74.07%                | 14.81%     | 1.91         | 0.06    |
| Family and Social History | 85.56%              | 79.75%                | 6.25%      | 1.03         | 0.30    |

\* Z score and P-values reports two sample t-test (two tailed) on the proportional difference between original cypher query evaluation questions and paraphrased questions. Significance at 0.01 level and no statistical significant differences are identified.

# Supplementary Figures



**Supplementary Figure 1:** AIPatient KG’s distribution of patients in each major diagnostic category. We also restrict the current analysis scope to patients aged 18 years and above and exclude cases of parturition, elective procedures, severe car accidents, or instances where the patient lost the ability to communicate or interact with medical care personnel. These cases are excluded because they involve clinical complexities that fall outside the intended focus of AIPatient’s patient-provider interaction model. Patients unable to communicate effectively are incapable of providing disease-related information to medical professionals. Similarly, including patients under 18 would introduce unique clinical and developmental complexities requiring a separate, pediatric-focused model. Cases like parturition, elective procedures, and severe car accidents also involve highly specialized care and specific clinical pathways that add complexity beyond the system’s current scope.



**Supplementary Figure 2:** Annotation Interface with Doccano. The MDs used the Doccano platform for NER annotation [51]. MDs were provided with predefined categories (e.g., symptoms, medical history, allergies) and marked relevant portions of the text that corresponded to these categories.

| Low Score                                | Big Five Traits                                                                      | High Score                                    |
|------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------|
| Practical, conventional, prefers routine | <b>Openness</b><br>imagination, feelings, actions, ideas                             | Curious, wide range of interests, independent |
| Impulsive, careless, disorganized        | <b>Conscientiousness</b><br>competence, self-discipline, thoughtfulness, goal-driven | Hardworking, dependable, organized            |
| Quiet, reserved, withdrawn               | <b>Extroversion</b><br>sociability, assertiveness, emotional expression              | Outgoing, warm, seeks adventure               |
| Critical, uncooperative, suspicious      | <b>Agreeableness</b><br>cooperative, trustworthy, good-natured                       | Helpful, trusting, empathetic                 |
| Calm, even-tempered, secure              | <b>Neuroticism</b><br>tendency towards unstable emotions                             | Anxious, unhappy, prone to negative emotions  |

**Supplementary Figure 3:** Big Five Personality describes human personality using five broad domains, each with a spectrum from low to high scores. To simulate personality profiles within the AIPatient system, we generated 32 unique profiles by randomly assigning either a high or low score for each of the five personality dimensions. Each trait was independently assigned, leading to 32 unique combinations.